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Electronic Companion to "When Front-Stage Innovation Overloads Back-Stage Workers"

(Authors' names are not included for peer review)

Abstract. This electronic companion provides full proofs, robustness analyses, and calibration details for the main paper.

Key words: electronic companion

1. Full Proofs

We restate each result for convenience and provide complete proofs. The model equations are:

$$D = (1 + aT) \cdot F^b, \tag{1}$$

$$G = D - K \cdot h(W), \quad h(W) = (1-p) + pW, \tag{2}$$

$$\frac{dK}{dt} = r\,[G]^+\,W - \delta K + u, \tag{3}$$

$$\frac{dW}{dt} = ([-G]^+ + f)(1-W) - (\mu + \sigma\,[G]^+)\,W, \tag{4}$$

$$\frac{dT}{dt} = \beta\,[-G]^+\,(1-T) - \eta\beta\,[G]^+\,T - \lambda T. \tag{5}$$

Customer lifetime value is denoted CLV throughout; starred variables (K^*, W^*, T^*) denote steady-state equilibrium values.

1.1. State-Space Invariance

LEMMA 1 **(Forward Invariance)**. *The set* $\Omega = \{(K,W,T) : K > 0,\ W \in [0,1],\ T \in [0,1]\}$ *is forward-invariant under the flow of the ordinary differential equation (ODE) system* (3)–(5) *for all* $t \geq 0$*, provided* $u > 0$*,* $f \geq 0$*, and all parameters are non-negative.*

A t each boundary, the vector field points into Ω: $\dot{W}|_{W=0} = [-G]^+ + f \geq 0$; $\dot{W}|_{W=1} = -(\mu + \sigma[G]^+) \leq 0$; $\dot{T}|_{T=0} = \beta[-G]^+ \geq 0$; $\dot{T}|_{T=1} = -(\eta\beta[G]^+ + \lambda) \leq 0$; $\dot{K}|_{K=0} = r[G]^+ W + u > 0$. Forward invariance follows by the standard barrier argument. □

REMARK 1 (PIECEWISE-SMOOTH DYNAMICS). The system (3)–(5) is piecewise-smooth, with a non-smooth gradient at the switching manifold $\mathcal{M} = \{G = 0\}$. Existence and uniqueness of solutions follow from Filippov's (1988) framework, provided trajectories cross $\mathcal{M}$ transversally; numerical verification at both equilibria confirms this under baseline parameters. The implicit function theorem arguments underpinning the analytical results are applied at surplus equilibria where $G^* = -S^* < 0$, so differentiability is guaranteed by the smooth structure of the ODE within $\{G < 0\}$; smoothing robustness is verified in §1.8.

REMARK 2 (COMPUTABILITY OF F_{crit}). The implicit balance condition $K^*h(W^*) = (1 + aT^*)F^b + S^*$ rearranges to $F = (K^*h(W^*)/(1 + aT^*))^{1/b}$, but this is self-referential since W^* and T^* depend on F. The computable form evaluates h and the demand term at the boundary $S^* \to 0$ where $T^* \to 0$, yielding the closed-form threshold $F_{\text{crit}} = (K^*h(f/(f+\mu)))^{1/b}$.

1.2. Equilibrium Existence and Uniqueness

Before proving the main theorems, we establish that the surplus equilibrium invoked by Theorems 1–3 exists and is unique. The equilibrium expressions derived in §4.1 of the main paper are conditional on a surplus $S^* > 0$, but S^* itself depends on the equilibrium values W^* and T^* through $S^* = K^*h(W^*) - D(T^*, F)$, creating a fixed-point problem. We resolve this by reducing the system to a single implicit equation in S^* and establishing that it has exactly one positive solution.

LEMMA 2 **(Surplus Equilibrium Existence; Uniqueness under C2)**. *For any* $F < F_{\text{crit}}$*, a surplus equilibrium* (K^*, W^*, T^*) *with* $S^* > 0$ *exists. Under Regularity Condition C2 (single-crossing condition C2-SC:* $\beta(f + \mu) < \lambda$*; analytic, see Step 3 above), this equilibrium is unique.*

A t surplus equilibrium, $K^* = u/\delta$. The equilibrium conditions yield $W^*(S) = (S+f)/(S+f+\mu)$ and $T^*(S) = \beta S/(\beta S+\lambda)$, both continuous and strictly increasing in surplus S. Define the surplus residual $\Phi(S) = (u/\delta)[(1-p) + pW^*(S)] - (1+aT^*(S))F^b - S$. Then $\Phi(0) > 0$ when $F < F_{\text{crit}}$, and $\Phi(S) \to -\infty$ as $S \to \infty$. Existence follows by IVT. For uniqueness,

$\Phi'(S) = \tau_1(S) - \tau_2(S) - 1$ where $\tau_1, \tau_2 > 0$ are decreasing. Under C2-SC ($\beta(f+\mu) < \lambda$), the equation $\Phi'(S) = 0$ has at most one solution (a decreasing function equals an increasing function), so Φ has exactly one positive root. C2-SC holds for all $f \in [0, 0.10]$: $\beta(f+\mu) \leq 0.018 < 0.025 = \lambda$. □

REMARK 3 (LOW-TRUST INTERIOR ATTRACTOR). A second fixed point exists for large F, with $T^*{=}0$ and $W^*_{\text{low}} = f/(f+\mu+\sigma G^*)$. The Jacobian has strictly negative eigenvalues (verified numerically for $F \in [1.0, 1.5]$). This interior attractor, jointly with the surplus equilibrium of Lemma 2, establishes the bistability invoked by Theorem 3.

LEMMA 3 **(Forward-Boundedness of Trajectories)**. *For any finite F and any initial condition $(K_0, W_0, T_0) \in \mathbb{R}_{>0} \times [0,1] \times [0,1]$, the solution satisfies $K(t) \leq \max(K_0, C(F)/\delta)$ for all $t \geq 0$, where $C(F) = r(1+a)F^b + u$. Combined with $W(t) \in [0,1]$ and $T(t) \in [0,1]$ (both preserved by the ODE structure), the state space is forward-invariant and bounded.*

U nder overload ($G > 0$), we bound $G^+ = D(T,F) - K\,h(W) \leq D(T,F) \leq (1+a)F^b$ (since $T \leq 1$) and $W \leq 1$. Therefore

$$\begin{aligned}\dot{K} &= rG^+W - \delta K + u \\ &\leq r(1+a)F^b - \delta K + u = C(F) - \delta K.\end{aligned}$$

By Gronwall's lemma, $K(t) \leq \max(K_0, C(F)/\delta)$ for all $t \geq 0$. Under surplus ($G \leq 0$), $G^+ = 0$ and $\dot{K} = -\delta K + u \leq 0$ whenever $K \geq u/\delta$, so K is bounded above by $\max(K_0, u/\delta)$. In both regimes the upper bound $\max(K_0, C(F)/\delta)$ holds. At baseline parameters with $F = 1.2$, this gives $K_{\max} \approx 25.3$, far above the operating range $K^* = 1.67$. Since W and T are bounded to $[0,1]$ by construction, the full state space is forward-invariant and compact for all finite F. □

REMARK 4 (SCOPE OF GLOBAL CONVERGENCE). Lemma 3 ensures all trajectories remain in a bounded compact set, a necessary condition for LaSalle's invariance principle. Which attractor a trajectory reaches depends on its initial condition relative to the basin boundary (Figure 2, main paper). For $F \geq F_{\text{crit}}$, only the low-trust attractor is stable (Remark 3). A complete global Lyapunov proof is left to future work: the principal obstacle is the switching manifold $\mathcal{M} = \{G = 0\}$, at which the vector field is discontinuous and the standard LaSalle invariance argument does not apply directly. A complete proof would require either a Filippov-specific Lyapunov construction (Filippov 1988) or a smooth-approximation argument; both are non-trivial for three-dimensional piecewise-smooth systems and constitute a natural extension of the present framework.

REMARK 5 (MAINTAINED CONDITIONS). Four conditions are maintained throughout: (C1) $\Phi(0) > 0$ (i.e., $F < F_{\text{crit}}$; analytic via IVT); (C2) uniqueness of S^*, established analytically under C2-SC ($\beta(f+\mu) < \lambda$, which holds for all $f \in [0, 0.10]$ independent of b, p, σ; see Step 3 above); (C3) regularity $\Phi'(S^*) < 0$, a consequence of (C1)–(C2), required for the implicit function theorem in Theorems 1–2; and (C4) low-trust attractor existence $F^b > (u/\delta)\, h(W^*_{\text{low}})$, verified numerically (Jacobian eigenvalues across $F \in [1.0, 1.5]$).

REMARK 6 (BIFURCATION CLASSIFICATION AT F_{crit}). The transition at F_{crit} is a saddle-node (fold) bifurcation in the effective one-dimensional dynamics of the surplus variable S: the stable surplus equilibrium and the separating saddle coalesce and disappear as $F \nearrow F_{\text{crit}}$. The Jacobian of the full system at the surplus equilibrium has one eigenvalue $-\delta < 0$ (decoupled K-equation) and two eigenvalues from the (W, T) sub-system with strictly negative real parts throughout $F \in [0, F_{\text{crit}}]$, confirmed numerically (baseline: two complex-conjugate eigenvalues with real part ≈ -0.07; constant term of the characteristic polynomial $\approx 0.021 > 0$ at $F = 0.90$). No Hopf bifurcation occurs: both eigenvalues retain strictly negative real parts up to and including the limit $S^* \to 0$ (constant term $\to 0.004 > 0$ at F_{crit}). The zero eigenvalue of the full system at F_{crit} belongs to the one-dimensional manifold along which S^* vanishes, consistent with a saddle-node in the reduced dynamics.

1.3. Proof of Theorem 1 (Front-Stage Trap)

THEOREM 1 **(Front-Stage Trap, restated)**. *Suppose $b > 1$ and $p > 0$. Define the threshold $F_{\text{crit}} = \big((u/\delta)\, h(f/(f+\mu))\big)^{1/b}$. Then: (i) for $F < F_{\text{crit}}$, the unique surplus equilibrium exists with $T^* > 0$ and $\mathrm{CLV}^* > \mathrm{CLV}_{\text{low}}$; (ii) for $F \geq F_{\text{crit}}$, no surplus equilibrium exists; all forward trajectories remain bounded (Lemma 3) and converge numerically to the low-trust attractor with $\mathrm{CLV} \approx \mathrm{CLV}_{\text{low}}$.*

W e show that the surplus equilibrium exists if and only if $F < F_{\text{crit}}$, and that beyond this threshold the system is attracted to the low-trust fixed point.

Step 1: Surplus equilibrium requires $\Phi(0) > 0$. From Lemma 2, the surplus equilibrium exists if and only if $\Phi(0) = (u/\delta)\, h(f/(f+\mu)) - F^b > 0$, which rearranges to $F < F_{\text{crit}} = \big((u/\delta)\, h(f/(f+\mu))\big)^{1/b}$. (Here we use the fact that at $S^* = 0$, $T^* \to 0$ and $W^* \to f/(f+\mu)$.)

Step 2: Within the surplus regime, surplus decreases in F. At the surplus equilibrium, $K^* = u/\delta$ is independent of F. Differentiating the surplus residual, $dS^*/dF < 0$ for F near F_{crit} because the demand derivative $b(1 + aT^*)F^{b-1}$ dominates the indirect effects through W^* and T^*. As $F \to F_{\text{crit}}$, $S^* \to 0$.

Step 3: Beyond F_{crit}, overload drives collapse. For $F \geq F_{\text{crit}}$, $G > 0$, and from (5):

$$\frac{dT}{dt} = -T(\eta\beta G + \lambda) < 0 \quad \text{for all } T > 0.$$

Trust declines monotonically. Through Loop R2 ($W{\downarrow}\to h(W){\downarrow}\to G{\uparrow}$), the gap widens, accelerating trust decline. The convexity of demand in F ($\partial^2 D/\partial F^2 > 0$ for $b > 1$) ensures the gap widens super-linearly, making the collapse irreversible under the reinforcing loop R2. The system converges to the low-trust interior attractor (Remark 3).

Boundary conditions. When $b \leq 1$: demand is concave in F; the balancing loop B1 can compensate and the regime shift does not occur. When $p = 0$: $h(W) = 1$, severing Loop R2; the system may still experience trust decline through the direct overload channel but without the workforce-mediated amplification.

1.4. Proof of Theorem 2 (Good Jobs Buffer)

THEOREM 2 **(Good Jobs Buffer, restated)**. *Suppose $p > 0$ and the surplus equilibrium is unique and stable (Regularity Condition: $\Phi'(S^*) < 0$; Lemma 2). Then $\partial\,\mathrm{CLV}^*/\partial f > 0$ for all $f \geq 0$.*

W e show that each link in the chain $f \to W^* \to h(W^*) \to K_{\text{eff}} \to S^* \to T^* \to \mathrm{CLV}^*$ is strictly positive.

Link 1: $\partial W^*/\partial f > 0$. At the surplus equilibrium, from (4) with $[G]^+ = 0$:

$$\begin{aligned} 0 &= S^*(1 - W^*) + f(1 - W^*) - \mu W^* \\ &= (S^* + f)(1 - W^*) - \mu W^*. \end{aligned}$$

Solving: $W^* = (S^* + f)/(S^* + f + \mu)$. Differentiating: $\partial W^*/\partial f = \mu/(S^* + f + \mu)^2 > 0$. This is positive, decreasing in f (diminishing returns), and increasing in μ (higher natural decay amplifies the marginal benefit of investment).

Link 2: $\partial h/\partial W = p > 0$. By construction of $h(W) = (1 - p) + pW$.

Link 3: $\partial S^*/\partial h = K^* > 0$. Since $S^* = K^* h(W^*) - D(T^*, F)$ and $K^* = u/\delta > 0$.

Link 4: $\partial T^*/\partial S^* > 0$. From the equilibrium expression $T^* = \beta S^*/(\beta S^* + \lambda)$: $\partial T^*/\partial S^* = \beta\lambda/(\beta S^* + \lambda)^2 > 0$.

Link 5: $\partial\,\mathrm{CLV}/\partial T^* > 0$. By assumption on the CLV functional form (w and ρ both increasing in T).

Combining links via the chain rule:

$$\frac{\partial\,\mathrm{CLV}^*}{\partial f} = \underbrace{\frac{\partial\,\mathrm{CLV}}{\partial T^*}}_{>0} \cdot \underbrace{\frac{\partial T^*}{\partial S^*}}_{>0} \cdot \underbrace{K^*}_{>0} \cdot \underbrace{p}_{>0} \cdot \underbrace{\frac{\partial W^*}{\partial f}}_{>0} > 0.$$

Diminishing returns: $\partial^2\,\mathrm{CLV}^*/\partial f^2 < 0$ follows from the concavity of W^* in f (the $(S^* + f + \mu)^{-2}$ factor is decreasing in f) propagated through the chain.

General-equilibrium correction. The full equilibrium embeds a feedback loop $f \to W^* \to h \to S^* \to T^* \to D$, which partially offsets the direct surplus gain by raising demand through higher trust. To establish that this feedback does not reverse the sign, we apply the implicit function theorem directly.

Sign established analytically. The equilibrium condition $\Phi(S^*) = 0$ defines S^* implicitly as a function of f. Differentiating:

$$\frac{dS^*}{df} = -\frac{\partial\Phi/\partial f}{\Phi'(S^*)}.$$

Since $\partial\Phi/\partial f = (u/\delta)\,p\,\mu/(S^*+f+\mu)^2 > 0$ and $\Phi'(S^*) < 0$ (Regularity Condition C3), we have $dS^*/df > 0$ for all $f \geq 0$. The sign of $d\,\mathrm{CLV}^*/df$ is therefore analytically established; it does not depend on the magnitude of the trust-demand feedback.

Magnitude of the GE correction (numerically confirmed). The trust-demand feedback damps the surplus gain by a factor proportional to $aF^b\,\beta\lambda/(\beta S^* + \lambda)^2/|\Phi'(S^*)|$, which we refer to as the GE correction ratio. At baseline parameters this ratio is approximately 0.09, confirmed across 100 parameter configurations spanning the empirically relevant range. The correction dampens but does not reverse the direct effect; $d\,\mathrm{CLV}^*/df > 0$ throughout.

1.5. Proof of Theorem 3 (Burnout Cliff)

THEOREM 3 **(Burnout Cliff, restated)**. *Suppose $\sigma > 1$, $p > 0$, and $F_0 < F_{\mathrm{crit}}$ (a surplus equilibrium exists at F_0). Consider a firm at the surplus equilibrium that implements a step increase in F from F_0 to $F_0 + \Delta F$. The outcome is governed by two thresholds:*

(i) Safety threshold (permanent-collapse onset). *For*

$$\begin{aligned}\Delta F \;&\geq\; \Delta F_{\mathrm{collapse}} = F_{\mathrm{crit}}(f) - F_0 \\ &= \Big(\frac{u}{\delta}\,h\Big(\frac{f}{f+\mu}\Big)\Big)^{1/b} - F_0,\end{aligned}$$

no surplus equilibrium exists at the post-upgrade level; the low-trust attractor is the only locally stable state (Remark 3), and convergence to it is confirmed numerically across the studied initial conditions. For $\Delta F < \Delta F_{\text{collapse}}$*, a new surplus equilibrium exists at the post-upgrade level. Convergence to this equilibrium is basin-dependent; numerical verification confirms recovery from pre-shock equilibrium initial conditions across the parameter space studied (EC §EC.4.2).*

(ii) Speed diagnostic (instantaneous-overload onset). *A secondary threshold*

$$\Delta F_{\text{rapid}} = \frac{S_0}{b\,F_0^{b-1}(1 + aT^*)}$$

marks the boundary above which the service gap turns positive at $t = 0^+$*, initiating a rapid self-reinforcing collapse to the low-trust attractor. Because* $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ *at baseline (*0.40 *vs.* 0.31*), a firm upgrading by* $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ *collapses permanently but through a* slow *trajectory invisible on a standard fiscal-quarter horizon. The safety threshold is* $\Delta F_{\text{collapse}}$*;* ΔF_{rapid} *characterises collapse speed, not safety.*

Step 1: Two thresholds. The permanent-collapse threshold follows directly from Theorem 1: for $F_0 + \Delta F \geq F_{\text{crit}}$, equation (**??**) at $S = 0$ gives $\Phi(0) \leq 0$, so no surplus equilibrium exists. Therefore

$$\Delta F_{\text{collapse}} = F_{\text{crit}} - F_0 = \left(\frac{u}{\delta}\,h\!\left(\frac{f}{f+\mu}\right)\right)^{1/b} - F_0.$$

At baseline ($F_0 = 0.80$): $\Delta F_{\text{collapse}} = 1.1104 - 0.80 = 0.310$.

For $\Delta F < \Delta F_{\text{collapse}}$, the post-upgrade $F_0 + \Delta F < F_{\text{crit}}$, so a new surplus equilibrium exists (Lemma 2). Convergence to this equilibrium is basin-dependent; Remark 4 establishes boundedness, and numerical verification confirms convergence from pre-shock equilibrium initial conditions across the parameter configurations studied.

Step 2: Instantaneous-overload diagnostic. The first-order demand increment after the upgrade is $\Delta D \approx (1 + aT^*)\,b\,F_0^{b-1}\,\Delta F$. Setting $\Delta D = S_0$ (the pre-upgrade surplus):

$$\Delta F_{\text{rapid}} = \frac{S_0}{b\,F_0^{b-1}(1 + aT^*)}. \tag{6}$$

At baseline: $\Delta F_{\text{rapid}} = 0.401$. Since $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ (0.401 vs. 0.310), the interval $[\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ defines a *slow-collapse zone*: the surplus equilibrium is gone, but $G(t_0^+) < 0$, so collapse proceeds through a gradual erosion of surplus rather than immediate overload. For $\Delta F \geq \Delta F_{\text{rapid}}$, $G(t_0^+) > 0$ and collapse is rapid.

Steps 3–4: Well-being dynamics and timescales. Under overload ($G > 0$), the tipping threshold $W_{\text{tip}} = f/(f + \mu + \sigma G_0^+)$ marks the level below which depletion exceeds recovery. In the slow-collapse zone ($G_0^+ = 0$ at t_0^+), W_{tip} is crossed gradually as surplus erodes. Two timescales govern speed: $\tau_{\text{adapt}} = 1/r \approx 2.5$ mo (Loop B1) and $\tau_{\text{deplete}} = 1/(\mu + \sigma G_0^+) \in [1.5, 4]$ mo (Loop R3). Near $\Delta F_{\text{collapse}}$, both are comparable, so convergence to the low-trust attractor may require hundreds of months (explaining why simulated cliffs for low-f scenarios exceed the analytic threshold — a slow-convergence artefact, not a model error).

Step 5: Three-threshold characterisation and numerical verification. Starting from the surplus equilibrium (K^*, W^*, T^*) at F_0, a step upgrade ΔF produces the following structure. Let $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ be the saddle-node threshold (Theorem 1) and let $\Delta F_{\text{rapid}} = S_0/[bF_0^{b-1}(1 + aT^*)]$ be the instantaneous-overload threshold. At the baseline scenario ($f = 0.03$: $0.401 > 0.310$), $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ and they define three distinct zones:

- **Safe zone** ($\Delta F < \Delta F_{\text{collapse}}$): surplus equilibrium exists at $F_0 + \Delta F$; trajectories recover after transient overload.
- **Slow-collapse zone** ($\Delta F_{\text{collapse}} \leq \Delta F < \Delta F_{\text{rapid}}$): surplus equilibrium gone; $G(t_0^+) < 0$, so collapse proceeds gradually; convergence to the low-trust attractor may require hundreds of months near the boundary.
- **Rapid-collapse zone** ($\Delta F \geq \Delta F_{\text{rapid}}$): $G(t_0^+) > 0$ immediately; the death spiral begins from $t = 0^+$.

Note on ordering. The ordering $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ depends on f; it reverses at high f (e.g., $f = 0.10$: $\Delta F_{\text{collapse}} = 0.49 > \Delta F_{\text{rapid}} \approx 0.40$), where the slow-collapse zone disappears. Simulations ($t_{\text{end}} = 500$ months, collapse criterion $T(\infty) < 0.15$; the gap between the two attractors renders this threshold unambiguous) confirm collapse boundaries for three workforce-investment scenarios starting from their surplus equilibria at $F_0 = 0.80$:

For $f = 0.01$ the simulated cliff (0.27) exceeds $\Delta F_{\text{collapse}}$ (0.179): the surplus equilibrium disappears at $\Delta F = 0.179$, but the trajectory takes > 500 months to converge for $\Delta F \in [0.179, 0.26]$, reflecting the long transients of the slow-collapse zone. For $f = 0.03$, the simulated cliff (0.29) falls slightly *below* $\Delta F_{\text{collapse}}$ (0.310), consistent with basin-boundary effects: the trajectory escapes the surplus attractor's basin for $\Delta F \in [0.29, 0.31)$ even though the equilibrium formally exists.

Note on the two-threshold interpretation. The Burnout Cliff involves two distinct thresholds that are easily conflated. $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ is the *safety boundary*: the maximum upgrade size beyond which no surplus equilibrium exists and permanent collapse is certain. ΔF_{rapid} is a *diagnostic*: the upgrade size above which the service gap turns positive at $t = 0^+$, triggering an immediate death spiral rather than a slow erosion. Because $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ at the baseline scenario (0.40 vs. 0.31), a firm in the zone $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ is already in permanent-collapse territory but experiences only gradual deterioration, making the collapse invisible on a standard fiscal-quarter horizon. The correct safety threshold is $\Delta F_{\text{collapse}}$; ΔF_{rapid} characterises collapse speed, not safety. $\square$

1.6. Proof of Proposition 1 (Complementarity)

PROPOSITION 1 **(Digitalisation-Workforce Complementarity, restated).** $\partial K_{\text{eff}}/\partial W = pK > 0$ *and* $\partial^2 K_{\text{eff}}/\partial W \partial K = p > 0$. *Workforce well-being determines* $F_{\max}(W) = (K^* h(W))^{1/b}$, *which is strictly increasing in* W.

Part 1: Cross-partial. From $K_{\text{eff}} = K \cdot [(1-p) + pW]$: $\partial K_{\text{eff}}/\partial W = pK$ and $\partial^2 K_{\text{eff}}/\partial W \partial K = p$. Both are strictly positive for $p > 0$.

Part 2: $F_{\max}(W)$ **at the regime boundary.** $F_{\max}(W)$ is defined as the value of F at which the surplus equilibrium of Lemma 1 ceases to exist, i.e., at which $S^* = 0$. From Lemma 1, at $S^* = 0$:

$$T^*(S) = \frac{\beta S}{\beta S + \lambda} \xrightarrow{S \to 0} 0.$$

Therefore, substituting $T^* = 0$ into the F_{crit} formula from Theorem 1 (Eq. 1 at $S^* = 0$):

$$\begin{aligned} F_{\max}(W) &= \Big(K^* \cdot [(1-p) + pW]\Big)^{1/b} \\ &= (K^* \, h(W))^{1/b}. \end{aligned}$$

This boundary formula is exact at $S^* = 0$ because $T^* \to 0$ is a direct consequence of the equilibrium structure, not an approximation.

Full implicit differentiation at interior equilibria ($S^* > 0$)**.** At interior equilibria, $T^* > 0$ and $F_{\max}$ solves the fixed-point equation $(K^* h(W))/(1 + aT^*(W, F_{\max})) = F_{\max}^b$. Total differentiation with respect to W yields:

$$\frac{dF_{\max}}{dW} = \frac{1}{b} F_{\max}^{1-b} \frac{d}{dW}\left[\frac{K^* h(W)}{1 + aT^*}\right],$$

where

$$\frac{d}{dW}\left[\frac{K^* h(W)}{1 + aT^*}\right] = \frac{K^* p}{1 + aT^*} - \frac{K^* h(W) \, a}{(1 + aT^*)^2} \frac{dT^*}{dW}.$$

Since $T^*(S^*)$ is increasing in S^* and $dS^*/dW > 0$ (Theorem 2, Link 2–Link 3), the correction term $-K^*h(W)a\,(dT^*/dW)/(1+aT^*)^2$ is negative, meaning the interior-equilibrium slope $dF_{\max}/dW$ is somewhat smaller than the boundary-formula slope $(p/b)\,(K^*)^{1/b}[h(W)]^{1/b-1}$. However, the sign remains positive: $dF_{\max}/dW > 0$ for all $p > 0$, $K^* > 0$. The boundary formula therefore provides a conservative upper bound on the interior slope: the interior slope is smaller (by approximately 8% at baseline parameters), so the formula overstates how rapidly $F_{\max}$ increases with W at positive trust levels. $\square$

1.7. Proof of Corollary 5 (Budget-Constrained Trap)

L et $B_{\min} \equiv c_F \cdot F_{\text{crit}}$ denote the budget at which the all-F corner exactly attains the Trap threshold.

Case 1: $B \geq B_{\min}$. The all-F corner allocates $F = B/c_F \geq F_{\text{crit}}$, where Theorem 1 implies $\partial\,\text{CLV}/\partial F \leq 0$ (the surplus equilibrium has ceased to exist and the system is at or beyond the Trap threshold). Theorem 2 simultaneously guarantees $\partial\,\text{CLV}^*/\partial f > 0$ for all $f \geq 0$. Therefore, reallocating the marginal dollar from F to f produces a strict first-order gain in CLV. The all-F corner is strictly dominated and the optimum satisfies $f^* > 0$ and $F^* < B/c_F$.

Case 2: $B < B_{\min}$. The all-F corner allocates $F = B/c_F < F_{\text{crit}}$, so the Trap is not binding and a surplus equilibrium exists. Theorem 2 guarantees $\partial\,\text{CLV}^*/\partial f > 0$, so the all-$F$ corner ($f = 0$) forgoes a positive return to workforce investment.

Near-$B_{\min}$ sub-case (analytic). The per-dollar returns to F and f are continuous in F on $\{F < F_{\text{crit}}\}$: both follow from the explicit equilibrium formulae for $W^*(S)$, $T^*(S)$, and Theorems 1–2 via the implicit function theorem. At $B = B_{\min}$ (i.e., $F = F_{\text{crit}}$), Case 1 establishes $\partial\,\text{CLV}/\partial F = 0$ analytically while $\partial\,\text{CLV}^*/\partial f > 0$ (Theorem 2). By continuity, strict dominance of f over F per dollar persists throughout a neighbourhood $B \in (B_{\min} - \varepsilon, B_{\min})$ for some $\varepsilon > 0$.

Interior sub-case (computational). For $B \in [B_{\min}/2,\, B_{\min} - \varepsilon)$, strict dominance of f holds computationally across cost ratios $c_F/c_f \in [0.5, 5]$ (the range spanning realistic investment cost structures).

In Case 1 and near $B_{\min}$ in Case 2, $f^* > 0$ follows analytically; the interior of Case 2 is a computational finding. $\square$

1.8. Smoothing Robustness

The theoretical analysis uses the exact piecewise-smooth operators $[\cdot]^+ = \max(\cdot, 0)$. Table 1 verifies that replacing $[G]^+$ with a tanh-smoothed approximation $[G]^+_\varepsilon = G/2 + G\tanh(G/\varepsilon)/2$ across

Table 1 Sensitivity of F_{crit} to tanh smoothing bandwidth ε.

Smoothing ε	F_{crit}	Change vs. exact (%)
Exact ($\varepsilon = 0$)	1.110	–
$\varepsilon = 0.001$	1.106	-0.09
$\varepsilon = 0.005$	1.104	-0.27
$\varepsilon = 0.010$	1.100	-0.63
$\varepsilon = 0.050$	1.078	-2.62

Notes.

Table 2 Structural mapping between WCT (restricted) and Oliva–Sterman (2001).

WCT term	**O&S counterpart**	**Correspondence**
$r\,[G]^+\,W$	Learning from failures	Product of gap and workforce effort
δK	Depreciation	Constant-rate technology decay
$([-G]^+ + f)(1-W)$	Recovery + staffing	Surplus and investment raise effort; saturates at $W=1$
$(\mu + \sigma\,[G]^+)\,W$	Corner-cutting + fatigue	Overload and decay deplete effort; vanishes at $W=0$
Trust stock T	No O&S counterpart	Collapses to demand at $\beta \to \infty$, $\lambda = 0$

Notes.

bandwidths $\varepsilon \in \{0.001, 0.005, 0.01, 0.05\}$ shifts F_{crit} by less than 1% for practically relevant bandwidths ($\varepsilon \leq 0.01$), confirming that all three theorems are invariant to smoothing in this range.

2. Structural Mapping: WCT and Oliva–Sterman (2001)

The WCT model recovers the core dynamics of Oliva and Sterman (2001) as a special case under three parameter restrictions: (i) $p = 0$ (workforce does not affect K_{eff}, so $h(W) = 1$); (ii) $a = 0$ (no trust-demand feedback, so $D = F^b$); and (iii) F constant (no cross-domain dynamics). Under these settings, the trust ODE remains active but effectively decouples from demand; setting additionally $\beta \to \infty$ and $\lambda = 0$ collapses T into the demand function, recovering the O-S perceived-quality attractiveness mechanism. Table 2 provides the term-by-term correspondence.

The $(1 - W)$ saturation factor replaces O-S's nonlinear lookup table; qualitative behaviour is identical. The 2:1 overload asymmetry ($\sigma = 2$) transfers exactly and, with the three restrictions lifted, generates Loops R1, R2, and the cross-domain Front-Stage Trap.

3. Sensitivity Analysis

Table 3 reports how each theorem's qualitative result responds to $\pm 50\%$ perturbations of each parameter from its baseline value. For each perturbation, we recompute the relevant threshold (F_{crit}, $\partial\,\text{CLV}^*/\partial f$, or $\Delta F_{\text{collapse}}$) and record whether the theorem's qualitative prediction holds, weakens, strengthens, or disappears.

Table 3 Sensitivity of theorem predictions to $\pm 50\%$ parameter perturbations. †Slower sensing ($r\downarrow$) allows overload more time to erode W before the self-correction loop activates, steepening the cliff.

Parameter perturbation	Thm 1 (Trap)	Thm 2 (Buffer)	Thm 3 (Cliff)
b: 1.3 $\to$ 1.0	Disappears	Holds	Weakens
b: 1.3 $\to$ 1.6	Strengthens	Holds	Strengthens
a: 0.4 $\to$ 0.2	Weakens	Holds	Holds
a: 0.4 $\to$ 0.6	Strengthens	Holds	Holds
p: 0.5 $\to$ 0.25	Weakens	Weakens	Weakens
p: 0.5 $\to$ 0.75	Strengthens	Strengthens	Strengthens
σ: 2.0 $\to$ 1.0	Weakens	Holds	Weakens
σ: 2.0 $\to$ 3.0	Strengthens	Holds	Strengthens
η: 2.0 $\to$ 1.0	Weakens	Holds	Holds
η: 2.0 $\to$ 3.0	Strengthens	Holds	Holds
r: 0.4 $\to$ 0.2	Holds	Holds	Strengthens†
r: 0.4 $\to$ 0.8	Holds	Holds	Weakens
δ: 0.03 $\to$ 0.015	Weakens	Strengthens	Weakens
δ: 0.03 $\to$ 0.06	Strengthens	Weakens	Strengthens

Notes.

Table 4 Supplementary sensitivity: leading-indicator lag and direct-channel share across key parameter ranges.

Outcome	**Parameter varied**	**Low**	**Baseline**	**High**
Leading-indicator lag (months)†	$\beta \in \{0.06, 0.12, 0.24\}$, $\sigma = 2.0$	21	12	6
	$\sigma \in \{1.0, 2.0, 3.0\}$, $\beta = 0.12$	11	12	12
Indirect-channel share (%)	$p \in \{0.25, 0.50, 0.75\}$	<1	<1	<1

†Leading-indicator lag: months by which W leads T before T crosses the 0.10 crisis threshold. Scenario: step upgrade $\Delta F = 0.40$ from $F_0 = 0.80$ (above $\Delta F_{\text{collapse}} = 0.31$), no workforce investment, starting from the surplus equilibrium. Lower β (stickier customer trust) lengthens the warning window; σ variation contributes less than two months at fixed β. Indirect-channel share: fraction of CLV uplift from f via the sensing channel (p=0 counterfactual) at $F = 0.85$; values $< 1\%$ confirm the capacity-multiplier channel dominates. See §6.1 of the main paper.

Key observations. p is the most uniformly influential parameter: reducing it weakens all three theorems simultaneously. b is critical for Theorem 1 (at $b = 1$ the Trap disappears); σ for Theorem 3 (at $\sigma = 1$ bistability vanishes). No single perturbation kills all three results.

4. Calibration Protocol

4.1. Parameter Sources

All time units are months. Rate parameters are expressed in months^{-1}; implied half-lives are $\ln(2)/\theta$. The calibration follows the partial-model testing and pattern-oriented fitting protocol of Homer (1993) and satisfies the validity criteria of Barlas (1996). Three parameters warrant additional discussion beyond Table 5; the remaining seven are fully documented there.

Demand super-linearity ($b = 1.3$). A 10% front-stage increase raises back-stage demand by approximately 13%. The Trap requires only $b > 1$; sensitivity in Table 3.

Table 5 Parameter calibration summary.

Parameter	Base	$\tau_{1/2}$	Basis	Sensitivity
b (demand super-lin.)	1.30	–	Bell et al. (2018); Acimovic and Graves (2015)	Trap requires $b > 1$; robust
a (trust–demand)	0.40	–	McKnight et al. (2002); Sirdeshmukh et al. (2002)	Weakens Trap at a=0.2; holds
p (complementarity)	0.50	–	Illustrative; Krusell et al. (2000)	All 3 weaken at p=0.25
σ (overload asym.)	2.00	–	Oliva and Sterman (2001)	Cliff disappears at σ=1
η (trust erosion asym.)	2.00	–	Sirdeshmukh et al. (2002)	Weakens Trap at η=1
r (sensing-adapt.)	0.40	1.7 mo	Tucker (2007); Argote and Epple (1990)	Robust
δ (depreciation)	0.03	23 mo	Illustrative	Robust ($\pm 50\%$)
β (trust-update)	0.12	5.8 mo	Illustrative	Primary lag driver (EC Tab. EC.4)
μ (nat. W decay)	0.05	14 mo	Demerouti et al. (2001); Dubé et al. (2020)	Robust
λ (nat. T decay)	0.025	28 mo	Illustrative	Robust

$\tau_{1/2}$: implied half-life $= \ln(2)/\theta$.

Workforce-technology complementarity ($p = 0.5$). A burned-out workforce ($W = 0$) delivers 50% of design capacity (Krusell et al. 2000). All results hold for any $p \in (0, 1)$.

Overload-recovery asymmetry ($\sigma = 2.0$). Magnitude comparator from Oliva and Sterman (2001); the Cliff disappears only at $\sigma = 1$ (Table 3).

4.2. Simulation Initial Conditions

Table 6 reports the initial conditions and control parameters for each figure. All simulations use the baseline model parameters from Table 2 of the main paper. Initial conditions are chosen so that each simulation begins in the surplus regime ($K_{\text{eff}} > D$), representing a retailer with adequate but not excessive back-stage capacity.

Basin-of-attraction map. Grid: $W_0 \in \{0.08, 0.10, \ldots, 0.80\}$, $F \in \{0.70, 0.75, \ldots, 1.95\}$ (37×26 trajectories). Doubled-grid validation (73×51) shifts the separatrix by at most 0.0075 W_0-units. Burnout Cliff collapse criterion: $T(t{=}500) < 0.15$.

4.3. CLV Functional Form

We model CLV using the geometric-series structure of Gupta et al. (2004): $\text{CLV} = w(T)/(1 - \rho(T))$, where $\rho(T)$ is an annual retention probability and $w(T)$ an annualised per-customer margin. We adopt the specification $\rho(T) = 0.80 + 0.18T$ and $w(T) = 80(1 + 0.25\max(T - 0.15, 0))$. The retention endpoints ($\rho(0) = 0.80$, $\rho(1) = 0.98$) match the 70–90% annual range in Gupta et al. (2004); £80 is an illustrative annual margin for UK omnichannel grocery. The model's analytical results require only that CLV is increasing in T, which holds for any $\rho'(T) > 0$, $w'(T) \geq 0$.

The kink in $w(T)$ at $T = 0.15$ does not affect any analytical result: Theorem 2 requires only $\partial\,\text{CLV}/\partial T^* > 0$, which holds everywhere since $\partial\,\text{CLV}/\partial T^* \geq w(T^*)\rho'(T^*)/(1-\rho(T^*))^2 > 0$ even

Table 6 Simulation initial conditions and controls by figure.

Fig.	(K_0, W_0, T_0)	F_0	u	f	$t_{\max}$	Notes
2	balance†	varies	0.05	0.03	500	37×26 grid; CLV thr. = 600
3	Analytical: $F_{\max}(W) = (K^* h(W))^{1/b}$; $K^* = u/\delta$					Boundary formula
4	$(1.60, 0.55, 0.50)$	0.80	0.05	varies	120	Three scenarios (A,B,C)
5	(K^*, W^*, T^*)	0.80	—	0.01–0.10	500	61-pt ΔF sweep
EC	eq. IC at F=0.85	0.85	0.05	0.00–0.20	120	Study 3: Thm 2 decomp.; p=0 counterfactual confirms structural-zero

†Balance manifold: $K_0 h(W_0) = (1+a{\cdot}0.5)F^b$, T_0=0.50. Grid: 37×26 (962 pts). Collapse: $T(500)<0.15$.

at the kink. A softplus replacement produces CLV values within 0.5% and qualitatively identical outputs.

5. Empirical Appendix

This appendix supplements the empirical analysis in §5 of the main paper.

US data sources. ACSI scores (0–100, `theacsi.org`); Glassdoor ratings (1–5, min 5 reviews/firm-year); SEC EDGAR 10-K financials via XBRL API; FRED series ECOMPCTSA (e-commerce share); BLS JOLTS (turnover/wages).

EU data sources. Trustpilot (1–5, complaint-biased); Glassdoor (1–5); Yahoo Finance (`yfinance`); UK ONS internet sales (J4MC); Eurostat (`isoc_ec_eseln2`).

Variables. K=CapEx/Revenue; W=Glassdoor overall; T=ACSI or Trustpilot (rescaled: $(s-1)\times 25$); $K\times W$=standardised interaction; Assets rank=within-year percentile (currency-neutral).

US merge. EDGAR backbone (276 obs, 17 firms, 2008–2024); left-join ACSI, Glassdoor, macro. Complete: 131 obs (47%). No duplicate keys.

EU merge. yfinance backbone (50 obs, 11 firms; 4 duplicate rows dropped). Trustpilot/Glassdoor merged with 1:1 validation. Complete: 20 obs (40%). Fiscal-year offsets $\leq$3 months (Tesco Feb, Next Jan, ABF Sep, H&M Nov).

US robustness. (i) Between-effects: $K\times W = +1.95$, p=0.41; (ii) Mundlak CRE; (iii) Lagged $K_{t-1}\times W_{t-1}$; (iv) COVID exclusion; (v) Leave-one-out; (vi) Wild bootstrap (9,999 reps); (vii) CR3 with $\mathrm{df}=G-k=9$.

EU robustness. (i) Currency-neutral controls (assets rank); (ii) UK-only/Continental sub-samples; (iii) Leave-one-out: $K\times W \in [-0.04, +0.82]$; (iv) Wild bootstrap: p=0.62; (v) CR2, $t(10)$: p=0.045; (vi) Imbens–Kolesar $\mathrm{df}_{\mathrm{eff}}$=7.2, p=0.055.

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